Dinusanth Surendran

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EXPERIENCE

Software Developer — *Jan 2026 – Present*
RBC Capital Markets — *Toronto, ON*

- Engineering within **Quantitative Technology Services** (Canadian Central Risk Book & Electronic Market Making team) using Spring Boot and Java; supporting quantitative models and financial data infrastructure for capital markets
- Authored deployment automation scripts and consolidated parallel project instances, cutting total execution time from **1 instance/min (unbounded)** to **<2.5 min total** by parallelizing **all 7** instances into a single optimized run
- Built an **end-to-end automated** deployment pipeline, automating SSH-based deployment of scripts, configs, and Spring Boot services across repositories; reduced manual deployment time by 30 minutes per release.

Software Engineer Intern — *Jul 2025 – Feb 2026*
LB Connect — *Remote*

- **Promoted to codebase gatekeeper** within 2 months; approved all backend PRs and managed production releases, maintaining a **<5% bug regression rate** post-release
- Partnered directly with CTO on sprint planning and system architecture, driving decisions across a fast-moving production codebase in PHP

Automation Analyst Co-op — *May – Dec 2025*
RBC — *Toronto, ON · Hybrid*

- Built and deployed a backend pipeline converting **1,000+ messy financial records/week** into structured CSVs, enabling **3× faster audit reviews** by eliminating parsing failures
- Rewrote legacy Python scripts into production-grade systems; reduced manual QA effort by **60%** via CI/CD integration and enterprise-grade automated testing
- Optimized Python, SQL, and Bash scripts for performance across enterprise financial workflows; contributed to Agile bi-weekly deployment cycles

Founder & Executive Director — *Feb 2023 – Present*
Youth Volunteer Engagement Network (Non-Profit) — *Remote*

- Founded organization from the ground up, drafting **100+ founding and governance documents** and scaling to **28 active members** through structured outreach and partnership programs
- Designing and building a full-stack website and CMS to manage member onboarding, event registration, and internal process workflows end-to-end

PROJECTS

Quantitative Backtesting Research Framework — *Jul 2025 – Present*
Python, NumPy, pandas, Plotly

- Refactored ETF/equity backtesting engine into a **modular, production-grade research framework** with plug-and-play strategy classes, config-driven runs, and structured experiment tracking
- Implemented full risk analytics suite; **Sharpe, Sortino, Calmar ratios**, max drawdown, profit factor, win rate — validated on 5Y multi-factor ETF strategy (CAGR 5.56%, 153 trades, NAV $130,955)
- Engineered realistic **slippage/commission modeling** and vectorized NumPy computations for high-performance backtesting across large historical datasets

Enterprise Options Trading Platform — *2025*
Python, FastAPI, Streamlit, Plotly, Black-Scholes, Monte Carlo

- Built a full-stack derivatives pricing platform using **Black-Scholes, Binomial Tree, and Monte Carlo** simulation (with variance reduction) for European and American options
- Integrated real-time market data to compute and visualize **Greeks (Delta, Gamma, Vega, Theta, Rho)** and interactive 3D pricing surfaces; FastAPI backend with Streamlit frontend

FIFA World Cup - Predictor
Python, NumPy, pandas

- Built a Python World Cup 2026 match prediction system combining **Elo ratings, Dixon–Coles Poisson goal modeling, and 12+ calibrated signals** (form, H2H, squad value, injuries, coach, weather, altitude, travel/jet lag, market odds) into a CLI that forecasts 1X2 outcomes, xG, and secondary stats (shots, corners, cards) for all 48 nations.
- Integrated free and optional APIs (Transfermarkt, Open-Meteo, ESPN, The Odds API, international results) with **caching, walk-forward backtesting, and in-sample calibration** on finished matches to keep predictions realistic and auditable via per-factor breakdowns.

EDUCATION

McMaster University — *Hamilton, ON*
B.Tech, Automation Systems Engineering Technology — *Sep 2023 – Apr 2028*
Relevant Coursework: Instrumentation & Control, Embedded Systems, PLC Programming, CAD for Design

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, Bash, PHP, JavaScript
Quant / Finance: Black-Scholes, Monte Carlo, Binomial Tree, Greeks, Sharpe/Sortino/Calmar, vectorized NumPy/pandas
Backend & DevOps: Spring Boot, FastAPI, REST APIs, PostgreSQL, CI/CD, Docker, GitHub Actions, Linux
Certifications: Meta Python Programming (2025) · Automate the Boring Stuff with Python – Udemy (2025)